



CALCULATION CHECKING

PURPOSE

This practice lists items to be addressed in the systematic checking procedure for the purpose of establishing a uniform calculation checking philosophy and process for Fluor generated calculations, ensuring accuracy and consistency, and ensuring coordination with discipline drawings. This procedure will streamline the department checking operation and reduce checking costs, while still maintaining quality. The level of calculation checking is performed as instructed in the project specific Activity Plan.

The major cost savings will occur by the following:

- Minimizing duplication of effort in repetitive checking.
- Making certain that the requirements have been properly communicated and understood in each step of the drawing production process.

Emphasis is placed on the Engineer performing adequate self check and doing it right the first time, rather than relying on the Checkers to produce a quality product. The objective is to prepare calculations for which modifications as a result of checking will be minimal.

Because not all projects are alike, and sites vary, checking should not be restricted to only the items listed in this practice. Checklists attached to this practice or to the applicable discipline specific Work Instructions shall be used as a guide when checking calculations.

SCOPE

This practice describes the responsibilities of the Engineering Manager or his/her designee Lead Engineer or his/her designee, Engineer (when applicable), Calculation Originator, and Calculation Checker with respect to calculation checking. It included the following sections:

- APPLICATION
- RESPONSIBILITY
- REVISIONS
- ACCEPTANCE CRITERIA
- ATTACHMENTS

APPLICATION

This practice should be used by discipline personnel involved in the calculation checking process. It should also be used in conjunction with the project specific Activity Plan, associated Work Instructions and referenced Checklists. Checking shall be performed to meet the intention of the checking method identified on the Activity Plan.

Test





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RESPONSIBILITY

Some of the responsibility listed below may be either combined or designated depending on the size of the project or the operation of that Business Line.

Engineering Manager

The Engineer Manager or his/her designee is responsible for ensuring the checking procedure is in place on the project.

Lead Engineer

- Ensure that calculations are made when required, that calculation content is accurate and correct, that calculations are properly checked, and that calculations are recorded and filed for future retrieval (prior to project close-out) in accordance with the project Records Retention Plan.
- Establish a method of verification that the project-specific Activity Plan has been followed and implemented for each design item or group of items.
- Assign qualified Engineers to check calculations.

Design Engineer

- Know, understand, and implement the requirements of each assigned task as defined in the Activity Plan, and seek clarification as required from the appropriate source.
- Understand all necessary physical interfaces as documented by other disciplines layouts, 3D models, and relevant supplier data.
- Ensure that the calculation content is accurate and correct
- Provide the following information on the front page of a calculation:
 - Client Name
 - Project Name and Location
 - Contract/WBS
 - Calculation Number
 - Date of completion of calculation
 - Name and signature of Originator and Checker
 - Checking Level
 - Total pages (optional)
 - Professional Engineer's stamp (if applicable)
- Provide the following information in the first pages of the calculation:
 - Design assumptions to address uncertainties
 - Design methodology description



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- Applicable references, codes and standards.
- As a minimum provide the following on every page produced by the originator including computer print-outs:
 - Contract Number
 - Calculation number
 - Initials of originator and checker.
 - -Page number
- As a minimum provide the following on every page of reference information included in the calculation:
 - - Contract Number
 - - Calculation number
 - - Page number
- For calculations with computer analysis, include:
 - Hard copy or electronic copy (as directed by project) that will enable recreation of model in the future and essential output as defined by Lead Engineer.
 - Software title and revision used in the calculations
 - Graphics, where they might facilitate understanding
- Perform a self-check of calculations.

Engineering Checker

It is not the Checker's function to look for ways to improve the design. The goal is to achieve a workable design, and not necessarily a textbook solution. If the original design appears to be overly conservative, the Checker should consult with the Lead Engineer. The Lead Engineer should decide if rework would be cost-effective considering the total project impact.

- Verify that the design conclusions are acceptable.
- Do not request revision of the original calculations for small errors or changes that do not change the results. Instead, a clarifying statement should be added to the calculations.
- Do not prepare a separate set of calculations. This would result in two unchecked sets of calculations.

REVISIONS

Revisions to completed calculations will be made when the Lead Engineer considers that the assumptions utilized in the original calculations are no longer valid or that the original conditions have changed. The Original Engineer will make revisions, where possible. All revisions made to a set of completed calculations shall be clouded, initialed, and



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dated by the engineer making the modifications. Update the cover sheet to reflect the revision and state the reason for change.

ACCEPTANCE CRITERIA

The signature of the originator on the calculation cover sheet, calculations, and design sketches indicate compliance with this document.

ATTACHMENTS

CSA Checklists

Attachment 01	Structural Checklist - 31Mar2009
Attachment 02	HVAC Checklist - 31Mar2009
Attachment 03	Plumbing Checklist - 31Mar2009
Attachment 04 - 09	Not Used

Piping Checklists

Attachment 10 – 19	TBD
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Electrical Checklists

Attachment 20	Electrical – 02Apr2010
Attachment 21 – 29	Not Used

Control Systems Checklists

Attachment 30	Control Systems – 30Jul2010
Attachment 31 – 39	Not Used

Mechanical Checklists

Attachments 40-49	TBD
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DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

STRUCTURAL CALCULATIONS CHECKLIST

The list below specifies the items to be checked while reviewing the different types of listed documents. This list can be customized to suit the project needs by the Lead Engineer and issued as applicable.

Project Data

- Project Name and Location
- Client Name
- Project Number
- Calculation Number
- Date and Revision Number
- Checking Level
- Originator and Checker
- Pages Numbered
- Professional Engineer Stamp
- Computer File Index Sheet
- Table of Contents
- Description or Introduction
- Design Basis and Methodologies Project Criteria, Code requirements, etc...
- Assumptions and References
- Guidelines, Specifications, Reports, etc...
- Engineer of Record Observations of construction
- Conclusions

Design Sketches

- Location and Elevations
- Gravity System Layout
- Lateral System(s) Layout
- Structural Steel Shapes and Sizes
- Rebar Sizes and Spacing
- Anchor Bolt Diameter, Lengths, Projections and Locations
- Connection Details
- Foundation Sizes and Shapes
- Constructability Concerns
- Underground Interferences
- Latest Certified Dimensional Outline (CDO) of Equipment
- Shipping Limitations

Material Properties

- Structural Steel Properties
- Concrete Strengths
- Reinforcing Bars
- Welding Electrodes
- Connection Bolts
- Anchor Bolts



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Geotechnical Data

- Geotechnical Report Available
- Soil Bearing Capacity
- Soil Density
- Poisson's Ratio
- Subgrade Modulus
- Boring Locations and Logs
- Shear Modulus
- Dynamic Shear Modulus
- Bottom of Foundations
- Pile Type and Capacity
- Seismic Parameters
- Site Specific Investigation
- Frost Depth
- Water Table
- Lateral Pressure Coefficients
- Sliding Coefficient of Friction
- Backfill Properties
- Anticipated settlement

Basic Loads and Combinations

- Self Weight
- Fireproofing Loads
- Piping Loads
- Electrical Loads
- Mechanical Loads
- Vehicle Loads
- Maintenance (bundle pull) Loads
- Platform Loads
- Live Loads
- Snow and Rain Load
- Wind Load
- Seismic Load (Torsion)
- Test Load
- Erection Load
- Thermal Load
- Vibration Load
- Impact Load
- Blast Load
- Soil Vertical and Lateral Loads
- Service Load Combinations
- Factored Load Combinations



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Computer Analysis and Design

- Global Parameters
- Analysis methods (ASD, LRFD, etc.)
- Material Properties (f'_c , f_y , E , G , Density, etc.)
- Geometry (joint coordinates, TOC, TOS, etc.)
- Boundary Conditions (pinned, fixed, roller, springs etc.)
- Frame Members (sizes, orientation, end releases, offsets, cracked sections, etc.)
- Plates (geometry and aspect ratios, thickness, material, cracked, local axis, etc.)
- Solids (geometry and aspect ratio, material, local axis, etc.)
- Steel/Concrete Member Input Design Parameters (L_b , L_{comp} , K , C_m , etc.)
- Basic Load Cases
- Load Combinations
- Dynamic Input and Results (periods, mode shapes, mass participation, scale factors, etc.)
- Member Design Results (end forces/connection forces)
- Unity checks
- Reactions/Foundation Forces
- Deflections

Structural Steel Design

- Galvanized, Painted
- Fireproofing – shop or field, concrete or proprietary
- Member Sizes and Code Checks
- Connection Design (Bolts, Welds, Gusset Plates, Pass-Through Forces)
- Base Plates and Grout
- Crane Rails and Monorails
- Constructability
- Deflections and Tolerances
- Drift, Settlement
- Vibrations
- Torsion Effects
- Thermal Effects
- Flooring Type and Spans
- Seismic Provisions
 - Compact Shapes
 - Column Strengths
 - Column Splices
 - Special Moment Frame Requirements
 - Intermediate Moment Frame Requirements
 - Ordinary Moment Frame Requirements
 - Special Concentrically Braced Frame Requirements
 - Ordinary Concentrically Braced Frame Requirements
- Welding Requirements
- Bolting Requirements
- Field Inspection



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Reinforced Concrete Design

- Reinforcing (axial, flexural, shear, torsion)
- Minimum and Maximum Reinforcing
- Temperature Reinforcing
- Spacing of Reinforcing
- Side Bar (Skin) Reinforcing
- Deep beam and Corbel Reinforcing
- Crack Control
- Special Coatings
- Water Stops and Embeds
- Control or Expansion Joints
- Bar Lap Splices and Bar Development Lengths
- Dowel and Joint Congestion
- Seismic Detailing
- Foundation Design
 - Anchor Bolts,
 - Interference between Anchor Bolt Layout and Tie configuration
 - Soil Bearing
 - Overturning, Sliding
 - Bearing and Flexure
 - One Way Shear ,Two Way (Punching) Shear
 - Settlement
 - Buoyancy, Frost Uplift, Frost Depth

Vertical Vessel Foundations

- Piping Loads
- Platform Loads
- C.G. of Vessel
- Support Reactions from Mechanical
- Dynamic Analysis
- Footing shape for constructability
- Anchor Bolt Pre-tensioning
- Anchor Bolt Forces
- Anchor Bolt Embedment/Reinforcing
- Anchor Bolt Projection (chair height)
- Dowel Bars
- Top Steel Required
- Side Steel Required
- Soil Bearing/Pile Load
- Stability
- Sliding
- Wind Gust Effect Factor ($T > 1.0\text{sec}$) – Vortex Shedding



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Horizontal Vessel and Heat Exchangers Foundations

- Piping and Platform Loads
- C.G. of Vessel
- Uplift at one end while setting
- Stacked
- Thermal Forces
- Bundle Pull Forces and Access
- Friction Slide Plates
- T-Shape Pier
- Pier notch to accommodate bottom nozzle
- Anchor Bolt Forces
- Anchor Bolt Embedment/Reinforcing
- Pier and Footing Reinforcing
- Soil Bearing/ Pile Load
- Stability (Sliding, Overturning)

Pipe Supports

- Loads (Pipe Operating, Pipe Empty, Electrical Trays, Air Cooler Loads, Wind, Seismic, Snow, Ice, etc.)
- Access Platforms
- Anchor and Friction Forces
- Fireproofing limits and weight
- Flare Line Considerations (Height, Slope, Hydrotest, etc.)
- Air Cooler Attachment, Platforms and Header Pipe Supports
- Connections and Seismic Detailing
- Strut Collector Forces
- Deflections, Drift
- Welding Inspections
- Bolt Pretension Requirements
- Faying Surface Preparation
- Thermal effects (Expansion Joints, Brace Locations)
- Base Plates and Anchor Bolts
- Lifting Considerations (modules, trusses, etc.)

Precast Concrete Pipe Supports – in addition to the items above

- Standardized and Symmetrical Configurations
- Lifting Devices
- Minimum Cure Time before Handling
- Embeds (Avoid protrusions that prevent stacking)
- Special Strut Connections
- Foundation Interface (base plate, socket, etc.)



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Heater Foundations

- Vendor Loads
- Stiffness Distribution of Loads
- Anchor Bolt Design (Thermal Loads)
- Pier, Footing and Pile Design

Pump Foundations

- Vendor Data (Operating Speed(s), Reactions, Anchorage Forces)
- Soil Parameters
- $\frac{1}{2}$ Soil Bearing/Pile Capacity
- Mass Ratio
- Small Pumps < 5000 lbs. (or Centrifugal < 500Hp or Reciprocating < 200 Hp)
 - Hand Frequency Check
- Large Pumps > 5000 lbs. (or Centrifugal > 500Hp or Reciprocating > 200 Hp)
 - Dynamic Analysis (Dynamic Forces, Embedment Effects, Damping Variations, Soil Stiffness Variation, Vibration Amplitudes)
- Anchor Bolt, Grout and Minimum Reinforcing

Compressor Foundations – in addition to Pump Foundations items above

- Time History(s)
- Frequency Checks
- Coast Down/Start-up Limits

Tank Foundations

- Empty ,Operating and Hydrotest Cases
- Seismic, Wind and Snow Loads
- Content Specific Gravity
- Sloshing Effects
- Convective Period/Forces
- Impulsive Period/Forces
- Freeboard Height
- Uplift Conditions
- Anchored versus Unanchored
- Foundation Design (Slope, Leak Detection, Sumps, etc.)
- Settlement
- Flexible Attachments (Seismic flexibility, stairs attachment)

DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

HVAC CALCULATIONS CHECKLIST

The list below specifies the items to be checked while reviewing the different types of listed documents. This list can be customized to suit the project needs by the Lead Engineer and issued as applicable.

General

- Scope of calculations clearly defined
- Applicable codes, standards and specifications are referenced.
- Basis of calculations established from the following:
 - Project requirements checklist
 - Design criteria
 - Client's requirements
- Calculations properly formatted and numbered including cover sheet
- Charts, tables, and other references are clearly identified
- Status of interdisciplinary design information appropriate to calculations
- Calculations issued in accordance with project procedures
- Calculations performed in accordance with approved methods and include the following:
 - Design basis
 - References
 - Design premise / assumptions
 - Authorized computer program input / output
 - Calculations in accordance with Fluor Practices
 - Procedures and guidelines
 - Design / interface sketches

Cooling/Heating Calculations

- CFM/Cu.m.per minute quantities for rooms satisfy room sensible load and air changes per hour requirement for the specific application or design basis
- CFM/Cu.m.per minute quantities are adjusted for elevation above sea level
- CFM/Cu.m.per minute quantities are adjusted for duct leakage (5 percent)
- Safety factor (10 percent) has been added to cooling and heating load calculations
- Psychrometric chart or data has been prepared for each AHU or AC unit
- Outside air quantity satisfies ventilation, exhaust, and air pressurization requirements
- Insulation values used in the heat transmission coefficient for exterior surfaces (walls, roof, floor) match with actual callouts on architectural plans
- Heat transmission and shading coefficient for glass doors and windows match with callouts on architectural plans
- All heated/cooled spaces and rooms are included in the calculation
- All calculations shall indicate sensible, latent and total loads
- Correct Indoor outdoor design data used for temperature/humidity.
- Supply fan heat has been added to cooling coil load
- Return fan heat has been added to cooling coil load

DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

- For Clean Room space, air pressurization calculations have been made for every room
- For Clean Room space, FDA CGMP requirements have been followed for determining room air cleanliness classifications and air changes per hour circulation

Misc. HVAC Calculations

- Fan static pressure calculations made for each fan selection and a minimum 10% safety added
- Static head calculations made for each pump selection and a minimum 10% safety added
- All ventilation fans capacity (airflow) is calculated based on the air changes requirements established in the project design criteria and/or applicable codes.
- Ventilation louvers are sized based on total exhaust or intake air requirements and the manufacturer's charts for the free area versus static pressure loss across the louver (for the air intake louver, the air velocity shall not exceed the max recommended by manufacturer for rain water penetration)
- VAV box or zone reheat (electric or hot water) capacity is individually calculated based on the primary supply air temp., room design temp. and min. supply air to the room
- Boiler and chiller capacity calculation shall include the sum of all heating or cooling calculations for all systems served plus a minimum 10% safety
- Building mechanical systems for all buildings which provide space heating and/or space cooling shall be sized no greater than 150% of the design load as calculated unless required by the project for future loads
- Future additional loads shall be estimated in the equipment capacity if identified in the project design criteria
- Expansion tank, storage tank, chilled water loop volume calculations are included and done correctly.

DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

PLUMBING CALCULATIONS CHECKLIST

The list below specifies the items to be checked while reviewing the different types of listed documents. This list can be customized to suit the project needs by the Lead Engineer and issued as applicable

General

- Scope of calculations clearly defined
- Applicable codes, standards and specifications are referenced.
- Basis of calculations established from the following:
 - o Project requirements checklist
 - o Design criteria
 - o Client's requirements
- Calculations properly formatted including cover sheet.
- Charts, tables, and other references are clearly identified
- Status of interdiscipline design information appropriate to calculations
- Calculations numbered and issued in accordance with project procedures
- Calculations performed in accordance with approved methods and include the following:
 - o Design basis / References
 - o Design premise / assumptions
 - o Authorized computer program input / output
 - o Calculations in accordance with Fluor Practices
 - o Procedures and guidelines
 - o Design / interface sketches

Water Pressure Calculation

- Residual water pressure at the building obtained.
- Meter pressure losses and building height losses included
- Pressure losses thru backflow preventer and pressure reducing valve have been included
- Check developed length to the most remote outlet
- Check the required minimum pressure for the fixtures is accounted in calculation.
- For fittings, a safety factor has been incorporated in total developed length.
- Estimated friction loss per 100 ft / 30m of piping used are from correct table for correct material.

Gas Demand

- Gas demand (CFH/Cu. Meters per hour) and pressure at each gas appliance or equipment obtained based on equipment schedule or manufacturer specification
- Total connected loads (demand) at the building
- Gas line size at the building or at the meter
- Gas pressure at the building
- Longest run from the meter or building to remote regulator
- Design or distribution system pressure line



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

Water, Sanitary, Piping Sizing

- Number of fixture and type of fixture used are correct
- Fixture units used based on correct code/table
- Demand for indoor and yard sprinklers is included in water pipe size (if applicable)
- Make up water and condensate amount for HVAC and other equipment considered in calculations.

Roof drain Piping

- Correct data used for the amount of rain falls
- Applicable local or other codes requirement accounted in calculation.
- Correct duration of rainfall is used.
- All roof areas including under roof overhangs, and all exposed areas subject to rainstorm are considered.
- Charts, tables and references used for the pipe sizing are correct.



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

ELECTRICAL CALCULATIONS CHECKLIST

The list below specifies the items to be checked while reviewing the different types of listed documents. This list can be customized to suit the project needs by the Lead Engineer and issued as applicable.

Project Data

- Project Name and Location
- Client Name
- Project Number
- Calculation Number
- Date and Revision Number
- Checking Level
- Originator and Checker
- Pages Numbered
- Professional Engineer Stamp
- Computer File Index Sheet
- Table of Contents
- Description or Introduction
- Design Basis and Methodologies Project Criteria, Code requirements, etc...
- Assumptions and References
- Guidelines, Specifications, Reports, etc...
- Engineer of Record Observations of construction
- Conclusions

Electrical calculations developed and performed on a project may include without limitation:

- o Lighting illumination levels
- o Ground resistance
- o Voltage drop
- o Feeder sizing and derating
- o Cable pulling tensions
- o Fault current levels
- o Load flow
- o Load summary
- o Relay and protective device coordination
- o Raceway fill
- o Equipment sizing / ratings
- o Heat tracing
- o Transmission line
- o Power factor correction
- o Harmonic analysis and correction
- o Motor starting voltage drop and currents
- o Power system stability

All calculations will be checked by a person designated by the Lead Engineer or Designer. Checking will not be performed by the same person responsible for developing the calculations.



DOCUMENT PREPARATION AND CHECKING – CALCULATIONS

ELECTRICAL CALCULATIONS CHECKLIST

The checker will sign and date each calculation sheet or computer generated report in the appropriate space. If no "checked by" signature block exists, it will be noted above the signature.

The calculations and all revisions will be checked for the following:

- ☐ Project name and number
- ☐ Department name
- ☐ Sheet numbering
- ☐ Identification of type of calculation
- ☐ Signature and date of person performing calculations
- ☐ Assumptions clearly stated and valid
- ☐ Equations identified and source noted
- ☐ Reference material complete
- ☐ Math is correct
- ☐ Computer system and program identified
- ☐ Computer input data complete and correct



CALCULATION CHECKING – CONTROL SYSTEMS

1.0 PURPOSE

This practice establishes a method to be implemented for checking Control Systems engineering calculations performed within a Fluor contract. It applies to all projects and work assignments. This document is referenced as acceptance criteria in Fluor Practice 000.270.1910 – Work Instruction: Calculations – General. It is the responsibility of the Project Lead Engineer to ensure that this practice is used by an experienced checker.

2.0 SCOPE

This practice includes the requirements for checking the calculations for the commodities that a control systems engineer requisitions on a project. It also covers the preparation of information and data that a control systems engineer provides to other disciplines for their calculations.

The practice includes information about the following major topics:

- REFERENCES TO OTHER FLUOR DOCUMENTS
- RESPONSIBILITY
- INSTRUMENTATION CALCULATIONS COVERED
- CALCULATION PREPARATION
- CALCULATION CHECKING

3.0 APPLICATION

This practice is to be used by all Control Systems technical personnel involved in the preparation and checking of calculations. This practice shall be used at the time when calculations are checked and initialed per the Activity Plan. It is a requirement that calculations be initialed per the Activity Plan 000.270.F0072 whether the calculation is done by Fluor or by others.

4.0 REFERENCE

Following documents are available on KOL and can be used as reference documents for this subject:

Practice 000.270.F0072 – Activity Plan: Control Systems

Practice 000.270.1910 – Work Instruction: Calculations - General

Guideline 000.270.1018 - Instrument Design Criteria

Guideline 000.270.1101 - Temperature Instruments

Guideline 000.270.1208 - Job Data Book – Control Systems

Guideline 000.270.1301 - Flow Instruments



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Guideline 000.270.1401 – Level Instruments

Guideline 000.270.1410 – Instrument Level Sketches

Guideline 000.270.1502 – Restriction Orifices

Guideline 000.270.1552 – Relief Devices

Guideline 000.270.1753 – Instrument Power/Air Consumer List

Guideline 000.270.1852 – Instrument Air systems

4.1 RESPONSIBILITY

Lead Engineer: Establishes and maintain the project calculations in the job book. Ensures that the calculation is done in accordance with Client and Fluor requirements and that the checker is an experienced Engineer.

Commodity Engineer: Performs calculations where required for instrument selection or provides information to vendors for calculations and checks the vendor's calculations.

Checker: Experienced Engineer responsible for verifying that calculations are done per Fluor and/or Client guidelines and responsible for initialing the Activity Plan as "checked".

4.2 INSTRUMENT CALCULATIONS COVERED

Calculations are required for instrumentation listed in items A thru F below and have Engineering content. The responsibility for calculations may be given to a vendor but Fluor will check the calculations. Item G contains tabulations of loads that are used for sizing equipment procured by other disciplines. :

- A. Control Valves
- B. Relief Valves, Rupture Discs
- C. Orifice Plates (flow & restriction)
- D. Other Flow Meters
- E. Thermowell Strength (Vibration)
- F. Level Devices
- G. UPS Systems Electrical Load Calculations, Building Heat Load Calculations, Instrument Air Requirements
- H. Intrinsically Safe (IS) Loop

CALCULATION CHECKING – CONTROL SYSTEMS**4.3 CALCULATION PREPARATIONS:****A. CALCULATIONS BY THE VENDOR**

Typically, the calculation for sizing of instruments and valves is done by the vendors that are supplying the commodity. The vendors have their own customized computer programs appropriate for the sizing and for the selection of instruments from their line of manufacturing.

For the projects where the vendor is responsible for the calculations, the following procedure will be utilized by the commodity engineer.

1. All the required process information will be obtained from the process engineer. All data received from the process engineer will have appropriate signature and revision noted. Process Data originals will be stored in the Control Systems Job Data Book per 000.270.1208. This information will be entered in the instrument datasheet which will be squad checked by the process engineer. The datasheet will be provided to the vendor with procurement documents. The commodity engineer shall verify that the data sheet is complete in terms of process data required by the vendor for sizing of the instrument. Any assumptions about the process conditions as defined by the process engineer shall be mentioned in the datasheet. Any process data assumptions taken by the vendor shall be approved by the process engineer.
2. The Commodity engineer shall also ensure that parameters such as accuracy, repeatability, maximum range and turndown are reviewed with process and the client to ensure that commodities are not over specified. The cost implications on other disciplines can be significant e.g.
 - a) Straight line runs for flow metering can dramatically increase piping layout.
 - b) Oversized control valves can affect relief and blow-down cases leading to increased PSV sizes and flare loading.
 - c) Parallel control valves and flow-meters for unnecessary high turndown requirements can increase the cost significantly.

These parameters shall be confirmed with the vendor to ensure that the project is not paying for more accuracy than is needed.

3. The Commodity engineer shall ensure that the datasheet entries are related to the project design criteria and not to the default instrument brand used by the client. This mainly relates to accuracy and repeatability requirements but may include environmental specifications as well.
4. It is very important to maintain the revision number of the process datasheets received from the process engineer, such that any changes in the process data can be tracked throughout the project. This is to make sure that the selection of the instrument is based on the latest process data. The Commodity engineer will also be



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responsible for maintaining a log showing the instrument tag, the date and the revision number of process data that was received from the process engineer.

5. Commodity engineer will coordinate with the vendor such that at any time the vendor has the latest process data for sizing and selection of instrument. This will be true for the complete procurement cycle of the instrument, particularly when final selection and sizing of the instrument is being done by the vendor. This is achieved by very close coordination with the vendor, including having the vendor engineer who is in charge of calculations, in house as a team member of the task force for necessary duration.
6. Commodity engineer will become familiar with the vendor specific programs and will use them for checking the vendor calculations as necessary. The commodity engineer may use a program that he or she is familiar with, for sizing and selection of instrument, but this will only be as a back up or for double checking the vendor calculation and selection. This will not replace the vendor's submitted certified calculation document.

Any software program used for the calculation must be per project and client requirements and must be listed on the Fluor Software Directory.

7. For finalizing the selection of instrument, the Commodity engineer will arrange for a meeting between vendor, the process engineer, control systems engineer and preferably the client where the instrument selection will be finalized.
8. Vendor will submit the documents showing the calculations and selection of instrument and valves per project document submittal procedures. The Commodity engineer will check the documents and return them to the vendor with comments. All calculations, whether done by the vendor or by Fluor require signatures per the activity plan. All calculations shall be kept in the Control Systems Job Data Book per 000.270.1208.
9. At the end of approval cycle, the vendor will submit the final calculation for each tagged item that will be stamped "CERTIFIED" by the vendor. As a minimum this document will also contain the following information:
 - Contract Number
 - Tag Number/Description
 - Revision Date
 - Revision Number
 - Initials of person performing calculation
 - Checked By
 - Approved By

B. CALCULATION DONE BY FLUOR

On some projects where Fluor may have to play a bigger role in doing the calculations, the Commodity Engineer will obtain the appropriate sizing and selection program from the vendor and perform the calculations by following the steps as mentioned in the section A above. This will be done in very close coordination with the vendor's

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representative; however the calculation of record will be done by the vendor and submitted to Fluor. It shall be vendor's responsibility to select and provide the right instrument for the application. The Commodity Engineer will document that the vendor is in agreement with the selection of the instrument.

.It is a requirement that calculations be initialed per the Activity Plan 000.270.F0072 whether the calculation is done by Fluor or by others. All calculations shall be kept in the Control Systems Job Data Book per guideline 000.270.1208.

4.4 CALCULATION CHECKING

For checking the vendor's calculations, the checker shall verify that the program used for calculations is acceptable, the process datasheet that is used for the calculations is accurate and the latest revision and any assumptions taken by the vendor are acceptable and approved by the process engineer.

The checker shall verify that Commodity Engineer has followed the above mentioned steps.

For some long lead items, the initial process data given to the vendor is preliminary, so it is very important for the checker to make sure that the selection and sizing of the valves and instruments is based on the latest process data. Checker should audit the procedure that was followed to keep track of process data changes and investigate the system and method that was followed to make sure that vendor has the latest process data at any time.

For checking the selection of instrument, the checker will use his experience, knowledge and the list of documents in Section 4.0, Reference above.

A. CONTROL VALVES

Refer to Fluor guideline 000.270.1501 – Control Valves and Positioners

Calculations are required for all control valve applications. Checker shall check the following:

- 1, Verify that the software program used for the calculation is as per project and client requirements. Verify that the software is listed on the Fluor Software Directory.
2. Data sheets are consistent with Scope of Services and are based on Client's Standards, Fluor Standards, or Industry Standards as required for the project.
3. Verify that valves meet the piping specification including the end connection and flange finish requirements for the control valves.
4. Verify that the size and the type of valves is shown correctly on the P&ID's. Verify that the failure mode matches the process requirement. Verify that characteristic curve for the control valve is proper for the application.



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5. Verify that the valve shut off class is correct.
6. Verify that the control valve noise level does not exceed the project specifications.
7. Verify that the valve does not cavitate or flash. The valve sizing programs perform these calculations and warn if these conditions occur. In such situations proper equipment, trims etc. with the right metallurgy shall be used.
8. Checker shall check into possible valve plugging that could occur with a cage valve design or noise reducing trim design because of small particles in a dirty process fluid.
9. Verify that the selected actuator type and size is proper for the application and meets the specifications. Verify that the valve failure position is specified correctly.
10. Verify if the positioner is required and if the selected positioner meets the client specifications and requirements for the application.
11. The checker shall verify that the valve-sizing coefficient C_v is selected such that the valve provides control under normal flow conditions, as well as at minimum and maximum flow. Verify that the C_v for the minimum and maximum flows are within the Client's specification.
12. Verify that any special requirements for valve opening or closing times are met. Verify that the valves will meet the shut off pressure requirements.
13. Verify that the requirements for partial stroke testing of the valves are met.
14. For MOV's, verify that the torque of the motor is sufficient to meet any special closing requirement.
15. The checker shall pay special attention to the sizing and selection of the control valves whose failure is the determining case for sizing relief valves. The commodity engineer shall get a list of such valves from the process engineer and coordinate the C_v selected and C_v required for these valves and the corresponding bypass valves closely with the process engineer. Limiting the C_v requirements on the control valve may also result in adding a mechanical stop on a valve to prevent it from opening more than allowed. Checker shall verify this requirement.
16. Checker shall check that the following equipment is specified with the valve if it is required for operation or is required per project specifications.
 - a. Lubricant valve for packing.
 - b. Solenoid valve.
 - c. Volume booster.
 - d. Air set with gauge.
 - e. Lock up valve.
 - f. Transfer valve.

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- g. Volume tank.
- h. Limit switch.
- i. Travel stop.
- j. Valve position indicator.
- k. Hand wheel (process engineer to determine)
- l. Environmental Packing
- m. Fire Safe rating

B RELIEF VALVES

Refer to Fluor guideline 000.270.1552 – Relief Devices

Checker shall verify that the software program used for the calculation is as per project and client requirements and is listed in the Fluor Software Directory. Sizing of Safety Relief Valves is done by the API 520 Method or by ASME method. Sizing for two phase flow is done by the (OMEGA Method) per API 520. Sizing for Thermal Relief Valve calculations shall be by ASME Method. Sizing of Rupture Disk shall be by the API 520 Method.

Calculations are required for all relief valve applications. Checker shall check the following:

1. Process data

Verify that the process data used for the calculations is latest received from the Process Engineer, any assumptions taken by the vendor are agreed by process engineer, any special client specifications requirements have been followed in selecting the valve. Verify that approved two phase flow method has been used as required.

2. Material Requirements

Verify that all the metallurgy requirements are in agreement with the vessel trim specification or piping line class specifications, including the flange finish and end connection.

3. Failure cases

Verify that the worst failure case has been (Largest flow rate) used for sizing the relief valves.

4. P&ID's

Verify that the sizes and the type of valve relief selected is properly shown on the P&ID's, including the spare valve if required and selected.

5. Valve Set Pressure

Relief valves will generally be set at the equipment design pressure.

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Under relieving mode the pressure in the vessel will rise above the set pressure by an amount known as the accumulation. This is normally restricted to 10%, 3%, for boilers, and 20% for fire cases. Always check the appropriate code; for example, code may restrict the fire case to a maximum of 10%.

6. Inlet Losses

Generally relief valve inlet piping and isolation valves should be one size larger than the valve. Piping pressure losses between the equipment item and the relief valve (including velocity head losses in Tee's, vessel outlet losses, relief valve inlet reduces etc.) should not exceed 3% of the set pressure. The pressure loss should be calculated using the RATED capacity of the pressure relief valve. In order to meet these requirements, valve sub-headers must usually not be less than one size smaller than the main process line size.

Process engineer is responsible to check all inlet piping to ensure that the pressure drop does not exceed 3 percent of the set point. Once the detail design is complete (after piping finishes their spool drawings) the process engineers confirms that his assumption for the inlet piping were correct. If that is not the case, the process engineer may revise the process data and the control systems engineer may have to do the sizing calculation again for that valve.

When there is excessive inlet pipe pressure loss, remote sensing pilot-operated pressure relief valves should be considered.

7. Valve Back Pressure

For most valves the back pressure in the outlet piping should not exceed 10% of the gauge set point of the valve. In the case of valves with low set points which would require disproportionately large outlet piping, consider using balanced bellows relief valves for which the back pressure can rise to 50% of the set point. If pilot-operated valves have been installed in a relief system where the superimposed back pressure can exceed the inlet pressure of the relief valve, the main valve could open allowing reverse flow. A back flow preventer should be specified in the sensing line to prevent this.

8. Lifting Lever and Caps

Verify that the requirements for lifting lever and Caps are met as per specifications.

9. Thermal Relief Valves

Lines or equipment which can be left full of liquid under non-flow conditions and which can be blocked-in by closing off all the connections, must be provided with thermal relief valves to relieve pressure built up by heating and thermal expansion of the contained liquids.

In most instances where a thermal relief valve is required, a 3/4" x 1" valve shall be installed.

Where it is possible to have a very high heat input rate to the blocked-in liquid filled

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system the liquid relief rate can be calculated using the formula in API 521.

10. Rupture Disc

A rupture disc is a non-reclosing pressure relief device designed to give a virtually instantaneous unrestricted pressure relief opening to a closed system at a predetermined pressure. Its purpose is to provide overpressure protection to a system that may be subjected to excessive pressures caused by malfunction of mechanical equipment, run away chemical reactions and internal or external fires.

All the factors that play role in rupturing the disc become vital in determining the size and selection of the device. Following are the major factors and the checker shall ensure that these are considered for calculations and selection of device:

- a) Pressure margin: difference between normal operating pressure of a system and pressure rating of disc
- b) Disc environment: conditions that will be encountered on the process side and on the atmospheric side of the disc.
- c) Type of system pressure: whether the pressure is static or pulsating/surging
- d) System operating conditions: pressure, vacuum, alternating pressure and vacuum
- e) Temperature at disc location: since disc metal strength is variable with temperature, temperature is to be related to the pressure rating.
- f) Volume, viscosity, flow and characteristics of process media.

Checker shall ensure that the metallurgy used for the device is compatible with the corrosion and temperature characteristics of the process media. Ductile metals most commonly used are aluminum, stainless steel, inconel, nickel, silver, platinum, hastelloy tantalum and titanium.

Checker shall ensure that the rupture disc is non-fragmented if it is used in combination with relief valve. Verify that the relief valve is de-rated properly if it is used in combination with a rupture disc.

Checker shall verify that rupture disc pressure design tolerance in combination with the maximum allowable operating pressure does not exceed the maximum allowable pressure for the service.

C ORIFICE PLATES**1. FLOW ORIFICE**

Refer to Fluor guideline 000.270.1301 – Flow Instruments

Calculations are required for all orifice plate applications. Checker shall check the following items:

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- a. The units of measurement are as per project specifications.
- b. The beta ratio should fall between 0.3 and 0.7 or as specified in the client specifications.
- c. Pressure Drop
For gas and steam services the differential in inches of water column shall not exceed inlet pressure in PSIA.
A differential of 100 inches water shall be used, if practical.
- d. Normally the orifice plate material is stainless steel, but check that metallurgy of the plate is consistent with piping specification.
- e. The roughness of the orifice plate shall be as per project specifications and shall be compatible with the piping flanges.
- f. Normal flow shall generally be 70 percent to 80 percent of the maximum flow range. Maximum flow shall not be more than 90 percent of maximum range. Minimum flow shall not be less than 30 percent of maximum range.
- g. Round off maximum flow to nearest number that is a multiple of 10 if possible.
- h. .Separate pressure and temperature instruments shall be used for compensation, the pressure tap should be located upstream and the temperature tap should be located downstream of the orifice plate.

2. RESTRICTION ORIFICE

Refer to Fluor guideline 000.270.1502 – Restriction Orifices

Calculations are required for all restriction orifice plate applications. Checker shall check the following items:

- a. Bore should be calculated and rounded off to the nearest 1/8" size or maximum flow.
- b. Thickness of plate is determined by the differential pressure across it and shall be calculated. The thickness is ¼ inch minimum.
- c. Insure there is no cavitation across the restriction orifice. The pressure drop does not go below the vapor pressure of the liquid.

D. OTHER FLOW DEVICES

Refer to Fluor guideline 000.270.1301 – Flow Instruments

Selection of right instrument for an application depends upon the required level of accuracy, uncertainty and repeatability for the measured data. It is important for a checker to have a clear understanding of this requirement so that he can make sure that most appropriate instrument has been used for the application.

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Software used for flow devices other than orifice meters are vendor specific. The vendor is responsible for the final sizing calculations.

Long lead flow instruments for large diameter pipes are expensive and have to be ordered much earlier in the project to meet the delivery dates. Process data provided at the early stage of the project is subject to changes with the detail design development and should be kept track of. For such long lead items the checker shall confirm that the process data is being tracked very closely and that the instrument is being manufactured as per latest process data.

Calculations are required for all flow meter valve applications. Checker shall verify the following:

1. Check the input data for calculations against the instrument data sheet, including pipe size, pipe schedule, end connection, flange finish, viscosity, molecular weight, temperature, pressure,
2. The Maximum/Normal/Minimum flow rates and Reynolds numbers fit the specified instrument.
3. The type of transmitter specified is appropriate for the application, fluid type and fluid properties.
4. The application of the flow technology is acceptable.
5. The turn down ratio is acceptable.

E. THERMOWELL

The approved method of sizing thermowells is the ASME Performance Test Code(PTC) 19.3(1998) based on the J.W. Murdock research and referenced in Fluor guideline 000.270.1101 and PIP practices. SmartPlant Instrumentation(SPI) uses this method of calculation. This method assumes several parameters such as tip thickness, taper, and bore sizes that might be invalid for non-standard thermowell calculations. The Brock method should be used for non-standard thermowell calculations.

Calculations are required for all thermowell applications. The checker shall verify the following items.

1. Check the input data for calculations against the instrument data sheet, including pipe size, pipe schedule, end connection, density, viscosity, temperature, pressure, nominal size of sensing element, metallurgy, U length, and velocity.
2. The limitation on wake to natural frequency ratio(R) is to be set to 0.8.
3. Verify that the calculation has passed the following tests
 - a. Maximum external pressure calculation(Pmax)
 - b. Frequency Ratio criteria length calculation(L1)
 - c. Stress Safety factor length calculation(L2)

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For the thermowells that are modified to meet the vibration calculations, checker shall check the following:

1. If the thermowell length has been reduced, the minimum thermowell immersion length shall be the lesser of 1/3 the pipe diameter or 3” from the pipe wall. First reductions are to be to standard size thermowells (e.g., 12 to 9, 9 to 6, etc.). Where it is not possible to select a standard thermowell length or if the calculation continues to fail, reduce the length in decrements of 1/4” until it passes.
2. If the thermowell root diameter has been increased, make sure that modified thermowell can be inserted and installed in the provided opening.

F. LEVEL INSTRUMENTS

Refer to Fluor guideline 000.270.1401 – Level Instruments
000.270.1410 – Instrument Level Sketches

Calculations are required for all level instrument applications and shall be shown on level sketches. The checker shall check the following items.

1. For Differential Pressure Instrument the checker shall verify the following:
 - a. Check the vessel/tank condition; atmospheric or pressurized.
 - b. Verify process requirement (overall or interface) and level measurement range.
 - c. Verify process fluid conditions; temperature, viscosity, % solid, & density.
 - d. Check the type of level measurement; dry, wet, sealed.
 - e. For wet leg installation, verify that selected fill fluid is compatible with the process and can withstand both the process and environment temperature extremes.
 - f. Verify the transmitter location with respect to vessel level tap.
 - g. Check actual transmitter calibration range calculation
 - h. Verify from mechanical vessel drawings that the nozzle elevations and ratings match the data that is used in vessel sketches.
2. For the following type of level instruments, the checker shall verify that the sketches are done based on the approved information provided by the vendors and the calculations, if needed, are performed by the vendors.
 - a. Float/displacer type instrument
 - b. Radar type instrument
 - c. Ultrasonic type instrument
 - d. Capacitance type instrument
 - e. Tank gauging instrument
 - f. Gauge glass and Magnetic level gauges

G. UPS Systems, Heat Load Calculations, Instrument Air Requirements

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Refer to Fluor guideline 000.270.1753 – Instrument Power/Air Consumer List
000.270.1852 – Instrument Air systems

For these items there are no calculations to be performed by the commodity engineer; however the data provided by the engineer is used by other disciplines to do the sizing calculations and determine the size of the equipment that they procure for the project. For that reason, it becomes very important for the commodity engineer to maintain this provided information in an auditable tabular form. The commodity engineer shall maintain and keep good back up to validate the information that he entered in the table. The checker shall check this table and use following considerations:

1 UPS Systems Electrical Load:

The checker shall check the following items.

- a. Depending upon the client specification that governs the requirements for the battery backup for the UPS, there may be multiple UPS systems required. The battery back up requirement for the UPS for Fire and Gas systems can be as much as 24 hours, for communication systems can be in hours, and requirement for the rest of the control, monitoring and protective systems can be from 30 minutes to 60 minutes. It is important to segregate the UPS loads such that the each UPS is sized appropriately. Multiple UPS's will obviously have more space requirement also.
- b. Not all the connected loads are operating at the same time and that should be accounted when sizing the UPS. This is particularly true for redundant and triple redundant systems
- c. When counting for the loads the following should be accounted for:
 - All the field devices that are to be fed from UPS, This will include UPS for any required analyzers, UPS for any critical field instrument,
 - Check the other systems that may require the UPS power e.g. power requirement for Machine Monitoring Systems, Compressor Control systems, Burner management system, Fire and Gas system including the field sensors, H2S detection system, Continuous Emission Monitoring System etc.
 - UPS power required in future for the provided "future spares and spaces" in the rooms.

2 Building Heat Loads:

The checker shall check the following items:

- a. Heat generated (Load) for DCS and IPS systems will be provided by the manufacturers. Since the number and type of cards changes during the detail design, it is better to get the heat load numbers for all the components, calculate

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the heat load for each cabinet and keep track of it during detail design for any changes.

- b. Include the heat loads for all the other cabinets in the room, including any compressor control cabinets, vibration monitoring cabinets, relay cabinets and other interface cabinets, PLC cabinets, any cabinets in the room that are provided by packaged equipment vendors e.g. metering cabinets, analyzer cabinets, operator consoles, monitors including alarm monitors, work stations, training stations, printers, etc.
- c. Heat load generated by the UPS systems is significant. These numbers are provided by the UPS vendors. For redundant UPS systems, although only half of the total UPS will be running at any time, heat generated by the stand-by inverter and rectifier is very significant and should be accounted for.
- d. Heat loads 'to be added in future' should be estimated for all the spares and space provided.

3 Instrument Air Supply

The checker shall check the following items.

- a. Verify that all the air user instrument, valves, actuators have been accounted for.
- b. Verify that any purging requirements for the junction boxes, control panels or buildings etc have been accounted for.
- c. Verify that the air requirement for any vortex coolers is accounted for.
- d. Verify that any instrument air requirement for analyzers and analyzer shelters has been accounted for.
- e. Verify that any instrument air requirements for the laboratory have been accounted for.
- f. Verify that any requirements for the instrument and maintenance shops have been accounted for.

H. Intrinsically Safe (IS) Loops

Unless otherwise noted, the following section is based on IEC standards. Intrinsically Safe (IS) loops differ from most other forms of instrument calculations in that all the field loop equipment (up to the Safe Area IS Barrier), including the interconnecting cable as a group, is required to be within certain allowed electrical parameters depending on the type of loop and associated IS Barrier specification. This calculation is required for the input measurement and the output control side of all loops which use the IS (Exia, b or c) explosion prevention protection method. This loop entity approach is more complex and requires a good knowledge of the design protection requirements, being similar to safety instrumented system (SIS), safety integrity level (SIL) assessment calculations. As a



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typical type of loop uses the same make and basic model of equipment with associated IS Barrier, it is common practice to perform IS calculations for a typical application loop, and using the maximum length of interconnecting cable permissible for the different Gas Groups. When performing calculations of cable length based on NEC, if the electrical parameters of the cable used are unknown, the following values may be used:

Capacitance 60 pF/ft

Inductance 0.20 uH/ft

It is important not only to satisfy the IS calculation from a safety requirements perspective, but to also check that there is sufficient voltage to drive the field transmitter to its maximum output, which with 4-20mA loops may be more than 20mA, this may present a challenge where the instrument internal diagnostic faults are programmed to drive the signal to say >22mA, for this reason it is preferred to drive the signal downscale to <3.6mA on detected faults. Some IS solenoids may also present volt-drop problems depending on distance and cable conductor size. This IS calculation checking guideline excludes Fieldbus systems, and the manufactures guidelines for segment loading together with IEC 60079-27 Fieldbus Intrinsically Safe Concept (FISCO), should be consulted. The IS electrical parameter limitations are determined by the assigned Hazardous Area Gas Group electrical ignition energy (not ignition temperature), Gas Group IIC (IEC) or A/B (NEC) being Acetylene/Hydrogen is the most stringent due to their low ignition energy of around 20μJ. However, when assigning a Hazardous Area, apart from determining the Zones or Divisions for likely release, the representative area Gas Group is selected for ease of electrical ignition and ease of ignition temperature, i.e. Temperature Class, these will normally represent different gases or vapors for that process area. This is important for equipment selection, because apart from ensuring that the Hazardous Area equipment has an acceptable testing organization or body certification, approved by the local Regulatory Authority, the equipment also needs to satisfy both the Gas Group and Temperature Class. The equipment IS certificate with associated test report, needs to be read (prior to purchase) to confirm that the equipment is suitable for the intended design application, as with many certificates, there are usually some stated limitations (usually physical or environmental), which may preclude or limit the intended use.

For projects based on IEC electrical standards, refer to IEC 60079-11 to determine the calculation requirements.

For projects based on NEC electrical standards per ANSI/NFPA 70 Article 504, refer to ANSI/ISA –RP 12.06 to determine the calculation requirements.

If the project is based on OTHER electrical standards such as GOST, then refer to the Intrinsic Safety document to determine the calculation requirements.

Based on the Client's standards a defined calculation format may be required which needs to be reviewed for any project impact.

Calculations are required for all IS applications. Checker shall check the following:

1. Verify the following items on the typical IS loop drawing, of importance are the electrical parameters of working voltage (U), current (I), Power (P), inductance (L), capacitance (C) and resistance (R) for cable L/R ratio and system volt drop. The device inductance



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(L), and device capacitance (C) electrical parameters can be assumed to be zero and certification documents are not required for simple apparatus devices such as switches, thermocouples, Resistance Temperature Devices (RTDs), etc.

2. There are typical IS loop designs to cover for all application and equipment combinations used on the project, some may be from previous projects kept in a common IS loop database.
3. The correct instrument equipment make and model with certifying body IS certificate number, including type of cable used, is identified for the typical loop together with the associated electrical parameters.
4. The total of all field equipment electrical parameters including the single and multipair cables, all fall within the maximum limits stated on the IS Barrier located in the Safe Area (or in an approved enclosure). This check ensures that the required IS safety in loop design is achieved. Where more complex circuits are involved with the use of more than one power source or with non-linear characteristics, expert advice should be obtained, also consult IEC 60079-25 Intrinsically Safe Systems for examples.
5. The maximum safe working volt-drop allowed for each item of field equipment including cables, at maximum loop load is given, and the total checked against the available output voltage from the IS Barrier, at the minimum specified supply voltage to the IS Barrier (if applicable). This check ensures that the IS loop functional design is achieved at worst operating conditions.
6. Confirm that the IS equipment certificates satisfy the Hazardous Area Zone (ia, ib or ic), although where possible one should standardize on the worst case application of Zone 0 (IEC) and Division 1(NEC) with Exia equipment and also check the representative Gas Group and Temperature Class.
7. The longest IS cable lengths indicated on the cable schedule, fall within the maximum lengths stated on the typical IS loop drawings for the respective Hazardous Area Gas Group.
8. Any specific application design notes that may be taken from the equipment IS certificate and general notes such as the code of practice used for the project and IS segregation rules, are noted on the typical IS loop drawing.
9. Correct IS loop drawing revision and approval signatures.